A targeted search for fast radio bursts from a radio source with likely magnetar origin

Shiqian Zhao [1], Chengwei Liang [1], Rui Luo* [1], Juntao Bai [2], Shuangqiang Wang [2], Shi Dai [3], Weiyang Wang [4], Yao Chen [1] and Lawrence Toomey [3]

[1]Department of Astronomy, School of Physics and Materials Science, Guangzhou University, Guangzhou 510006, China
[2]Xinjiang Astronomical Observatory, Chinese Academy of Sciences, 150 Science 1-Street, Urumqi 830011, China
[3]CSIRO Space and Astronomy, PO Box 76, Epping, NSW 1710, Australia
[4]School of Astronomy and Space Science, University of Chinese Academy of Sciences; Beijing 100049, China

Abstract

Fast radio bursts (FRBs) are transient astrophysical events characterized by millisecond-duration, bright radio signals emitted from extragalactic sources. Understanding the origin and nature of FRBs remains a significant challenge in modern astrophysics. Up to now, what we are aware of is that at least a subset of them originate from magnetars, which are extremely strong magnetized compact objects in the universe. In 2023-2025, using the FAST and Parkes radio telescopes, we conducted a 96-hour targeted search for fast radio bursts from a radio source, VT 1137−0337, which has exhibited a series of multiwavelength characteristics of magnetar wind nebula similar to the persistent radio sources (PRS) of some active repeaters, such as FRB 20121102A and FRB 20190520B. After the data reduction using TRANSIENTX, we found some candidates suspected to be true signals, which were determined to be false positive candidates caused by the fluctuation of noise through further analysis, so the possibility of the selected candidates being real signals was excluded. Furthermore, based on the nondetected case at VT1137−0337, we have imposed strict limits on the flux density of any potential radio radiation that might come from this source.

Introduction

Fast radio bursts (FRBs) are bright millisecond-duration radio bursts of unknown physical origin observed at extragalactic distances (Lorimer et al., 2007; Petroff et al., 2019). In the early stage, theory predicted that FRBs could originate from magnetar activity such as large flares (Popov & Postnov, 2013). As evidence of FRBs from multiple observations of magnetars emerged, magnetars began to emerge as top-candidate ancestors of FRBs.

Located at R.A. = 11:37:06.19, Dec = −03:37:37.3 (J2000), VT 1137−0337 was originally discovered and reported by Dong & Hallinan (2023) at a luminosity distance of 121.6 Mpc. VT 1137−0337 exhibits a series of magnetar wind nebula multi-wavelength features similar to active repeaters such as FRB 20121102A (Marcote et al., 2017) and FRB 20190520B (Niu et al., 2022). If VT 1137−0337 were found to be a FRB repeater, it will strengthen the connection between magnetars and FRBs and provide opinions into the origin of FRBs.

Observation and Data Analysis

We have been conducting the targeted observations for VT 1137−0337 using the Five-hundred-meter Aperture Spherical Radio Telescope (FAST) and the Parkes 64-m radio telescope with a total observing time of 96 hours during the period of 2023-2025, aiming to search for fast radio bursts from this radio source. Our FRB search is performed using the CPU-based software package TRANSIENTX (Men & Barr, 2024). In the pipeline based on TRANSIENTX, each data block will be normalized before searching, and then radio frequency interference (RFI) mitigation will be performed. After this, the pipeline would go through de-dispersion, matched filtering, and clustering until sifting the candidates. To search for FRB event at any possible dispersion measure (DM), we keep conducting de-dispersion uniformly from 0 to 2000 $\mathrm{pc\,cm^{-3}}$ in our pipeline. This ensures that we do not miss any candidate signals with unconscious bias during the search.

So far, we have completed the processing of all the observational data, and no significant radio signals were detected. Furthermore, from the observations of Parkes and FAST, we screened out two candidate single pulses that might be like signals from the generated candidates, with DM values within the estimated range, which are shown in Figure 1. All the observation durations of VT 1137−0337 are shown in Table 1. However, through further analysis, we ruled out the possibility that these candidates were real signals. They might be suspected candidates generated from fluctuations of noise. In the single pulse search using the TRANSIENTX, a lower S/N threshold and more severe RFI will increase the probability of TRANSIENTX false positive.

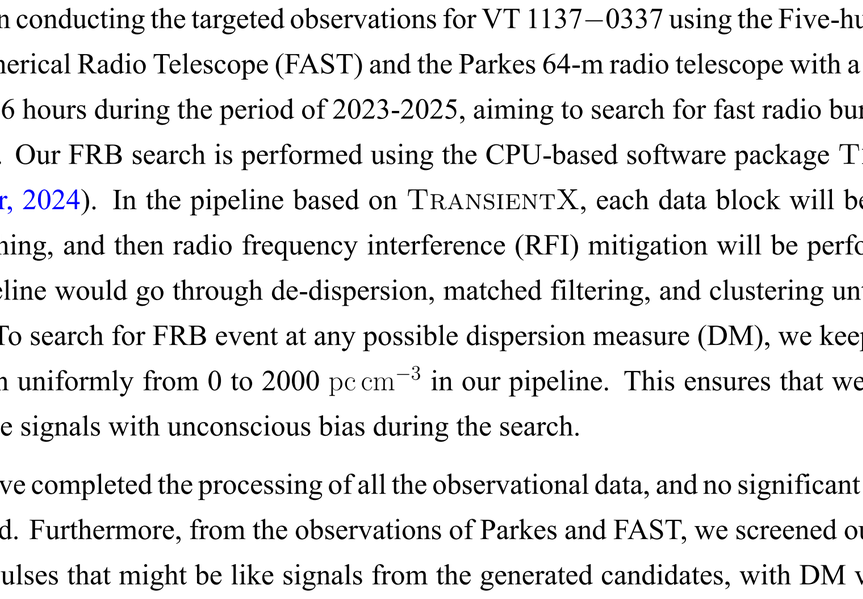


Figure 1. The suspected radio pulse signal candidates from VT1137−0337 were searched by TRANSIENTX[a]. The left panel shows the candidate found on 22 Jun 2024 by FAST, and the one on 23 Jun 2024 by Parkes is shown on the right. In each sub-plot, the pulse profile, the dynamic spectrum of the pulse, and basic meta-information about the pulse are shown.

[a]https://github.com/ypmen/TransientX

Table 1. The observations of VT 1137−0337 using Parkes and FAST radio telescopes.

Date	Length(s)	Telescope	Date	Length(s)	Telescope	Date	Length(s)	Telescope
2023/10/20	7200	Parkes	2024/08/19	10800	Parkes	2025/02/03	7200	Parkes
2023/11/07	7200	Parkes	2024/08/20	7200	Parkes	2025/02/16	7200	Parkes
2023/11/08	7200	Parkes	2024/08/24	7200	Parkes	2025/02/20	7200	Parkes
2023/11/18	7200	Parkes	2024/09/04	7200	Parkes	2025/02/06	7200	Parkes
2023/11/21	7200	Parkes	2024/09/13	5400	Parkes	2025/03/22	7200	Parkes
2023/11/22	7200	Parkes	2024/09/20	9000	Parkes	2025/03/26	3600	Parkes
2023/11/24	7200	Parkes	2024/10/07	7200	Parkes	2022/07/02	7200	FAST
2023/11/26	7200	Parkes	2024/10/12	7200	Parkes	2023/08/23	3600	FAST
2023/12/01	7200	Parkes	2024/11/02	7200	Parkes	2023/11/01	3600	FAST
2023/12/15	7200	Parkes	2024/11/12	7200	Parkes	2023/12/24	3600	FAST
2024/07/06	3600	Parkes	2024/11/22	7200	Parkes	2024/02/09	3600	FAST
2024/07/07	3600	Parkes	2024/11/30	7200	Parkes	2024/03/24	3600	FAST
2024/07/19	7200	Parkes	2024/12/07	7200	Parkes	2024/05/15	3600	FAST
2024/07/22	14400	Parkes	2024/12/17	7200	Parkes	2024/07/07	3600	FAST
2024/07/25	5400	Parkes	2025/01/04	7200	Parkes	2024/11/23	3600	FAST
2024/08/03	10800	Parkes	2025/01/11	7200	Parkes	2025/03/23	3600	FAST
2024/08/08	3600	Parkes	2025/01/21	7200	Parkes	2025/04/22	3600	FAST
2024/08/17	9000	Parkes	2025/02/01	7200	Parkes			
Total time: 84-h (Parkes); 12-h (FAST)								

In addition, we can determine the upper limit on the flux density for VT 1137−0337 based on the results from nondetection for the single pulse search. Referring to the radiometer equation and assuming that the minimum detectable S/N=9, we calculated the upper limit of the single-pulse peak flux density within the range of 0.1 to 100 milliseconds, as shown in Figure 2.

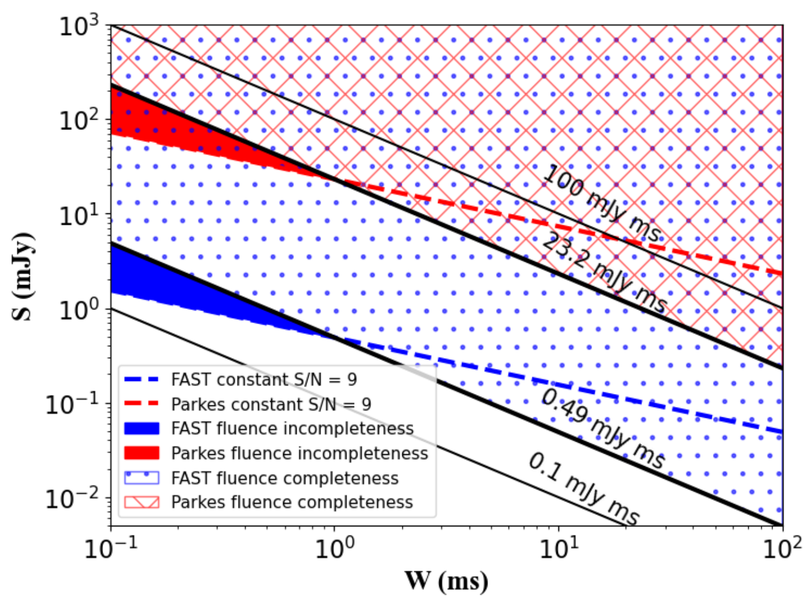


Figure 2. The detection capabilities of FAST and Parkes for FRB. In this plot, the dotted line indicates a constant signal-to-noise ratio, and the solid line indicates a constant flux. In the dotted areas, FAST maintains fluence completeness, whereas the gridded regions indicate Parkes' fluence completeness, and shaded areas indicate fluence incompleteness (blue: FAST, red: Parkes).

Discussion and Summary

In this work, we conducted long-term observations of VT1137−0337 using the Parkes and FAST telescopes, with the Parkes observing for 84 hours and the FAST observing for 12 hours. For the selected candidate pulses, we ruled out one by one the signals that originated from real celestial objects and were more likely false positives caused by fluctuations in noise. Although no significant radio signals analogous to FRBs were detected from VT 1137−0337, our results impose strict upper limits on the flux density of any potential radio radiation from this source, and the physical origins of PRS for some active FRB repeaters can be discussed. In addition, the radio luminance of VT 11370337 is relatively lower than that of FRB 20121102A and FRB 20190520B, which might imply its lower radio efficiency to produce bright FRB events like what has been detected from those two active repeaters. To better monitor possible burst events from this source, we call for multi-band observational investigative on VT 1137−0337, which may provide us with some potential clues related to the relationship between magnetars and fast radio bursts.

References

Dong, D. Z., & Hallinan, G. 2023, ApJ, 948, 119

Lorimer, D. R., Bailes, M., McLaughlin, M. A., Narkevic, D. J., & Crawford, F. 2007, Science, 318, 777

Marcote, B., Paragi, Z., Hessels, J. W. T., et al. 2017, ApJ, 834, L8

Men, Y., & Barr, E. 2024, A&A, 683, A183

Niu, C. H., Aggarwal, K., Li, D., et al. 2022, Nature, 606, 873

Petroff, E., Hessels, J. W. T., & Lorimer, D. R. 2019, A&A Rev., 27, 4

Popov, S. B., & Postnov, K. A. 2013, arXiv:1307.4924

cwliang@e.gzhu.edu.cn